9.3 Cross Sections

Lesson Introduction

Benchmark Focus	MA.912.GR.4.1 Identify the shapes of two-dimensional cross-sections of three-dimensional figures.	Learning Target	• I can identify the shapes of two-dimensional cross sections of three-dimensional figures.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• What are cross sections? • How can you determine the shape of the cross section of a three-dimensional figure? • How do you sketch the shape of a cross section of a three-dimensional figure?
Lesson Narrative	This lesson gives students a collaboration opportunity to determine the shape of cross sections of three-dimensional figures. Students make connections between the three-dimensional figures and their cross sections. The latter part of the lesson makes students identify and sketch the cross section. The lesson has a mix of collaboration as well as guided and independent work.

Component Summary	9.3.1 Warm-Up (~5 min.)	Finding a pattern in a sequence of images *(SE p. 65)*	9.3.3 Guided Instruction (15-20 min,)	Drawing the cross sections of figures on a grid *(SE p. 67)*
	9.3.2 Collaboration (10-15 min.)	Determining the shape of cross sections of prisms, cylinders, cones, and spheres *(SE p. 66)*	9.3.4 Your Turn! (5 min.)	Practice drawing cross sections of figures *(SE p. 69)*
	9.3.5 Wrap-Up (~5 min.)	Drawing the cross sections of figures on a grid *(SE p. 70)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Glossary: • Cross Section	• (Optional) Colored Pens/Pencils • (Optional) Straightedge	• Consider using manipulatives and videos that provide a visual demonstration of cross sections.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may incorrectly identify cross sections of a horizontal plane when compared to a vertical plane.
Students may not correctly apply the dimensions properly when sketching cross sections.</td><td>Consider providing additional help and/or scaffolding on 9.3.3 Guided Instruction questions 3 and 4, as these questions require specific dimensions to accurately draw the cross sections.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Three Reads
The questions in the 9.3.3 Guided Instruction and 9.3.4 Your Turn! give a perfect opportunity to execute the Three Reads strategy. Have students read the problems three times to be sure they fully comprehend what the questions are asking. Here is an example of three reads.

Read 1: Teacher reads the question aloud. Ask students to consider: What is this situation about?
Read 2: Class does choral read or partner read of the question stem. Ask students to consider: What are the quantities in the situation?
Read 3: Partner or choral read the problem stem orally one more time. Ask students to consider: What mathematical questions can we ask about the situation?</td><td>MA.K12.MTR.2.1: Multiple Representations
9.3.1 Warm-Up provides an opportunity to use multiple representations when describing a sequence. Students may draw all seven shapes, make a table of values, or write an algebraic equation to represent the sequence. Provide an opportunity for students to share their representations and discuss how each representation means the same thing.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The concept of cross sections is visual in nature. Use this lesson to differentiate for students who tend to think visually. Some students may struggle with conceptualizing cross sections. For a visual entry-point, use manipulatives and videos that demonstrate cross sections.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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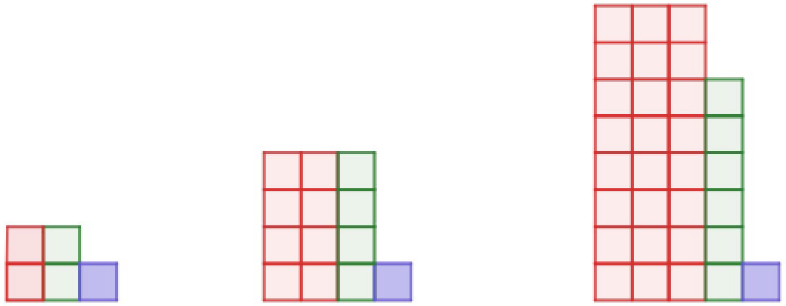
9.3.1 Warm-Up: Patterns

Component Overview: *Students are given a diagram and are asked to use patterns to determine the number of red squares in step 7.*

9.3.1 Warm-Up: Patterns

1. How many red squares are in the seventh pattern?



Answers will vary. The horizontal dimension of the red blocks is n, where n is the pattern number. The vertical dimension of the red blocks is 2^n. The total number of red squares for any pattern number, n, is $n(2^n)$. For the seventh pattern there will be $7(2^7) = 896$ red squares.

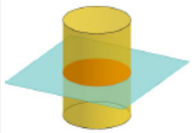
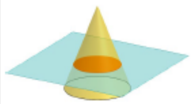
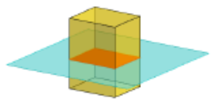
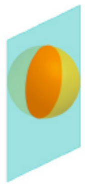
To highlight MA.K12.MTR.2.1 (Multiple Representations), some students could make a table of values to find the solution. Some students may also be persuaded to use a formula. Others may choose to draw all seven images. Allow students multiple pathways to find a solution to this problem. Provide an opportunity for students to share their representations and then make connections between them.

9.3.2 Collaboration: Cross Sections

Component Overview: *Students will examine the table of five three-dimensional figures and their vertical and horizontal cross sections. A Notice and Wonder will be completed to guide students through the process of learning about cross sections and how to identify the cross section of a three-dimensional figure. Students will complete a table with the three-dimensional figures by identifying the cross sections.*

9.3.2 Collaboration: Cross Sections

The table below shows the resulting cross section for each type of slice.

Figure	Slice	
	Horizontal Plane	**Vertical Plane**
Cylinder		
Cone		
Rectangular Prism		
Triangular Prism		
Sphere		

Use the following questions as prompts to guide student thinking and understanding of how cross sections are formed:

- *Why are the horizontal and vertical cross sections of the cylinder, cone, and triangular prism different?*
- *Why are the horizontal and vertical cross sections of the sphere and the rectangular prism the same?*

9.3.2 Collaboration: Cross Sections (continued)

1. What do you notice about the diagrams for horizontal slices?
   Answers will vary. I notice that the horizontal diagrams match the base of the three-dimensional object.

2. What do you notice about the diagrams for vertical slices?
   Answers will vary. I notice that the vertical diagrams make shapes that you might not see from looking at the 3-D shape.

3. Discuss with your partner whether or not the shape of a cross section will change if the height of the plane is changed. Summarize your discussion.
   Answers will vary. In vertical slices, the size changes but not the overall shape because the height of the figure is one of the slice's dimensions. In horizontal slices, the shape, nor size, of the cross section will not change when the height of the figure changes.

4. Complete the table by describing the shape of the cross section that is created when the figure is sliced by each.

Figure	Cross Section Shape	
	Horizontal Plane	**Vertical Plane**
Cylinder	*Circle*	*Rectangle*
Cone	*Circle*	*Triangle*
Rectangular Prism	*Rectangle*	*Rectangle*
Triangular Prism	*Triangle*	*Rectangle*
Sphere	*Circle*	*Circle*

The Notice and Wonder provides students with a discussion opportunity. Have students jot down their Notice and Wonder and allow students to discuss their observations verbally.

For a kinesthetic approach, allow students to move to other students in the classroom and give an opportunity to discuss and share their noticings and wonderings.

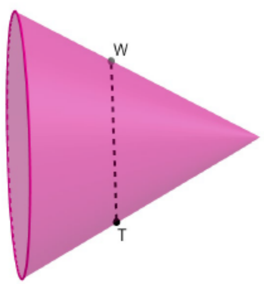
9.3.3 Guided Instruction: Drawing Cross Sections

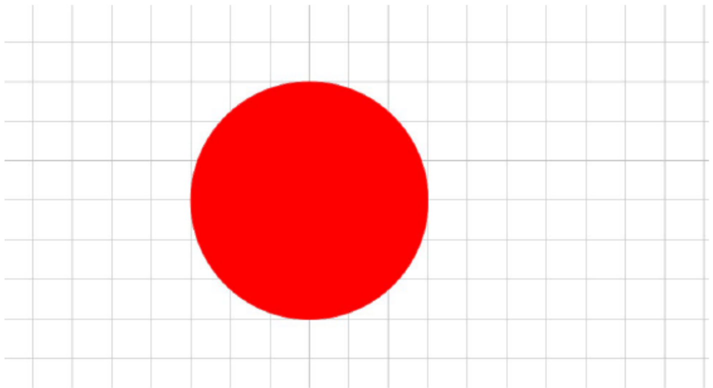
Component Overview: *This component of the lesson extends from 9.3.2 Collaboration by having students sketch the resulting cross sections on a grid when given the three-dimensional figure. Each question gives dimensions that students are expected to apply to sketch the cross section.*

9.3.3 Guided Instruction: Drawing Cross Sections

Three-dimensional figures can be oriented in different ways. Cross sections are typically described by their relation to the base of the three-dimensional figure. For example, cross sections are often described as parallel or perpendicular to the base of the figure.

1. A cone is shown with a cross section parallel to the base. Draw the cross section of the cone on the grid, given $WT = 6$ units.

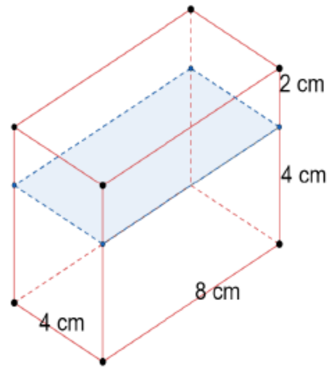


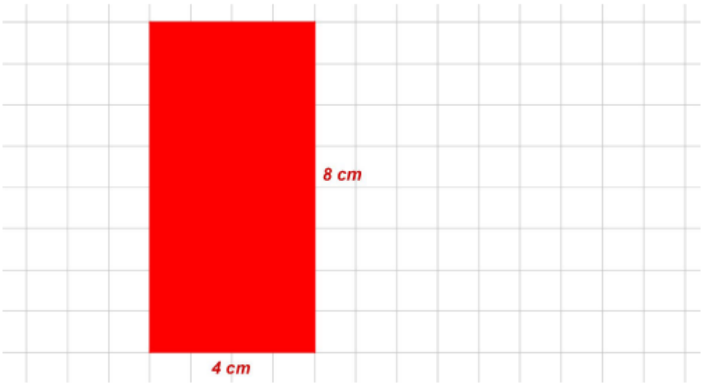


How would the vertical cross section change if the cone was standing on its base instead of its side?

9.3.3 Guided Instruction: Drawing Cross Sections (continued)

1. A rectangular prism is shown with a cross section parallel to the base. The dimensions are given in centimeters (cm). Draw the cross section on the grid and label its dimensions.

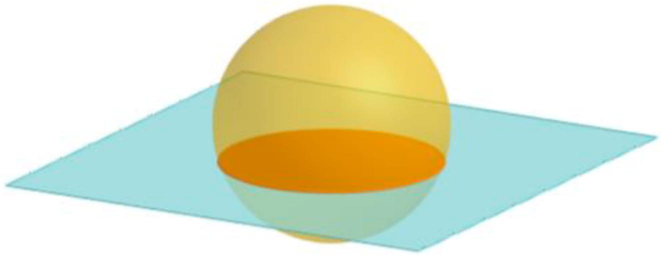




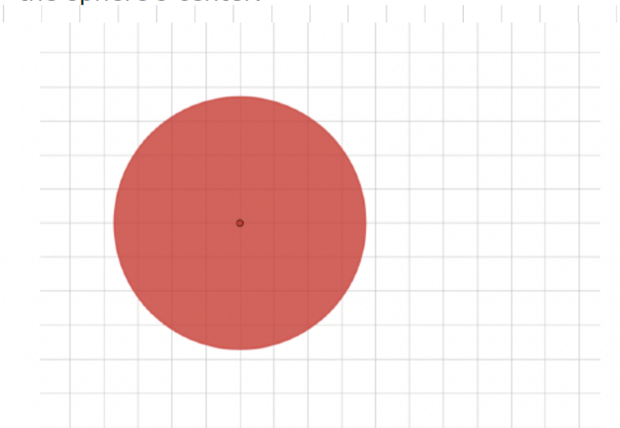
Be sure that students label the proper dimensions when graphing the cross sections to indicate what one unit of the grid represents.

9.3.3 Guided Instruction: Drawing Cross Sections (continued)

3. A sphere with radius of 4 inches (in.) with a horizontal cross section is shown.



Draw the cross section if it occurs $\frac{3}{8}$ of the distance from the sphere's center.



A cross section that is $\frac{3}{8}$ of the distance from the center of the sphere will be located $4 \cdot \frac{3}{8} = \frac{6}{4} = 1.5$ in. from the center of the sphere. Using Pythagorean Theorem, $1.5^2 + b^2 = 4^2$, the cross-section circle that is $\frac{3}{8}$ of the distance from the center of the sphere will have a radius of a distance of $\sqrt{13.75} \approx 3.708$ inches.

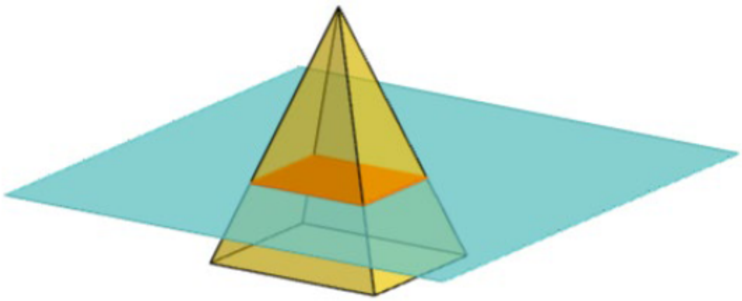
In the answer key, one unit of the grid represents 1 inch.

Students may require additional help and scaffolding for question 3. Consider asking the following when needed:

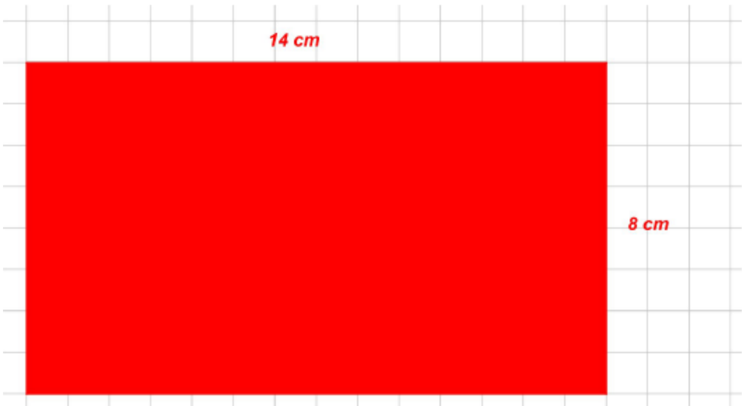
- *If the cross section is $\frac{3}{8}$ of the distance from the center, how far is it from the edge of the sphere?*
- *How can you find the radius of the cross-section circle that is scaled from the center of the sphere? (Multiply 4 by $\frac{5}{8}$.)*

9.3.3 Guided Instruction: Drawing Cross Sections (continued)

3. A rectangular pyramid is shown, where the dimensions of the base are 12 centimeters (cm) by 21 cm and the pyramid's height is 33 cm.



Draw the cross section parallel to the base that is located at a height of 11 cm.



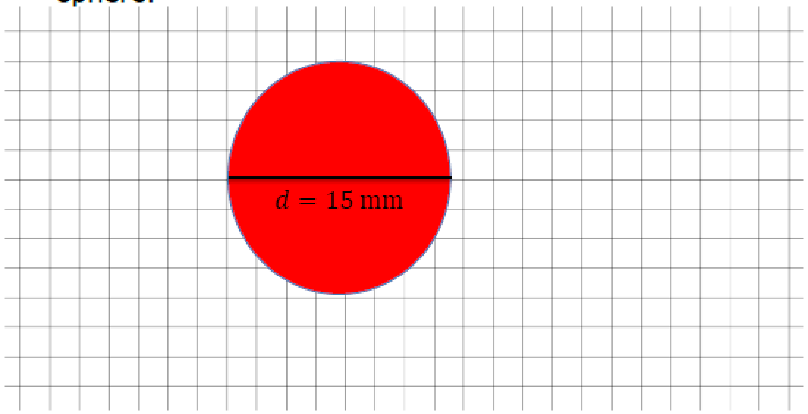
Provide opportunity for additional scaffolding as needed on question 4. Students will have to multiply the dimensions by $\frac{2}{3}$ because the cross section is $\frac{2}{3}$ the height of the pyramid (the perpendicular distance between the top of the pyramid to the bottom).

9.3.4 Your Turn!

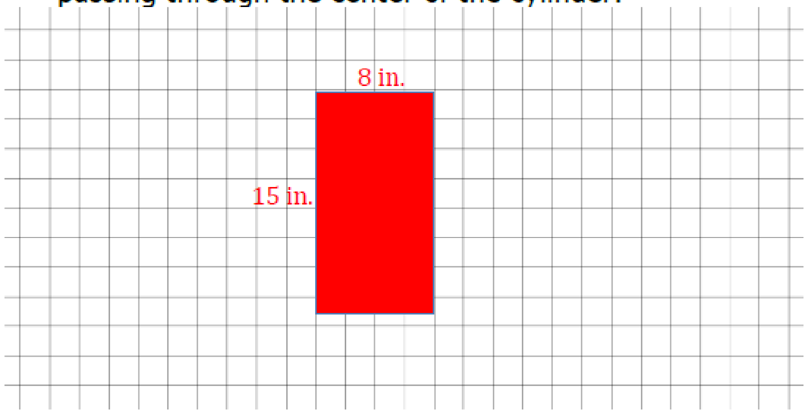
Component Overview: *Students practice drawing two-dimensional cross sections of three-dimensional figures on a grid.*

9.3.4 Your Turn!

1. Draw the cross section formed by the intersection of a sphere with a diameter of 15 millimeters and a perpendicular plane passing through the center of the sphere.



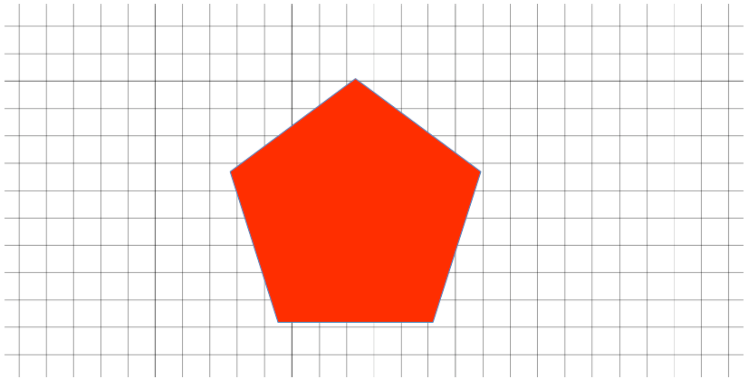
2. A cylinder has a radius of 4 inches (in.) and a height of 15 in. Draw the cross section formed by the intersection of a cylinder and a plane perpendicular to the base passing through the center of the cylinder.



Before sketching, have students identify what shape a cross section is of the intersection of a sphere and a perpendicular plane. For further scaffolding, review the term "perpendicular." Also, have students pay attention to the dimensions given so that the cross sections are sketched accurately. In the shown answer key for questions 1 and 2, one unit of the grid represents 2 millimeters. Allow students to make choices of what one unit represents for their drawings.

9.3.4 Your Turn!

3. Draw the cross section formed by the intersection of a pentagonal prism and a horizontal plane passing through the prism.



For differentiation, assign students one of the three questions and have a student become an "expert" on the question. Then, group students into groups of three, where each student was assigned a different question. Then have students share how to complete their questions.

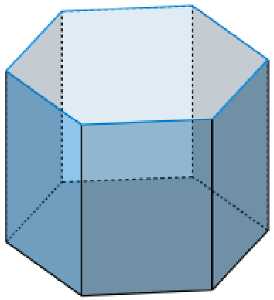
Question 3 has no given dimensions of the pentagonal prism. Correct answers will be any-size pentagon.

9.3.5 Wrap-Up: Ticket Out the Door

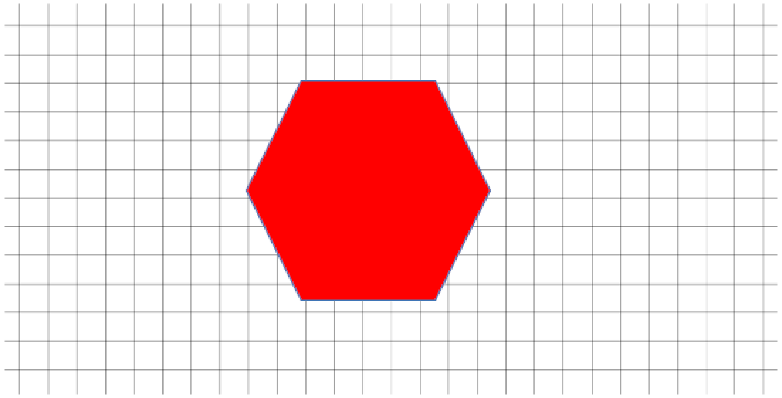
Component Overview: *Students complete a question where they are required to sketch a cross section that is horizontal to the plane and vertical to the plane.*

9.3.5 Wrap-Up: Ticket Out the Door

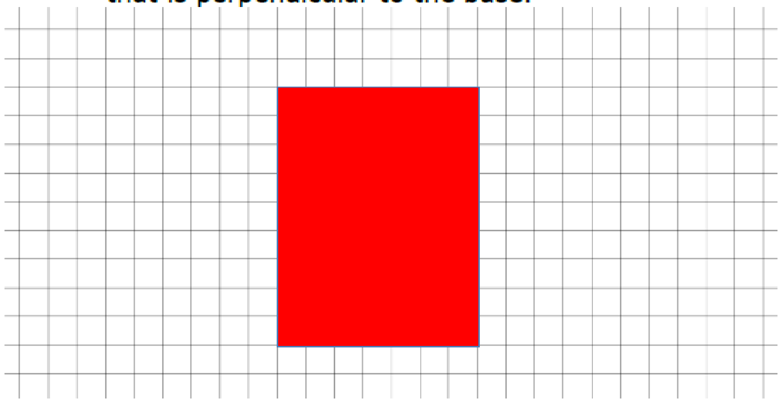
1. A hexagonal prism is shown.



   a. Draw the cross section that results from a plane that is parallel to the base.



   b. Draw the cross section that results from a plane that is perpendicular to the base.



Cross sections are important for Lesson 4 in Unit 9. Consider using the results of 9.3.5 Wrap-Up to form groups for 9.4.2 Activity.

Questions 1a and 1b have no given dimensions of the three-dimensional shapes. Correct answers will be any-size hexagon and rectangle. The shape of the rectangle will vary depending on where the plane passes through the shape.